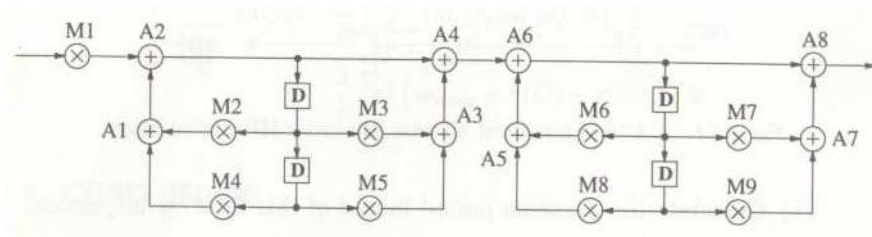


2026 Spring VLSI DSP Homework Assignment #4

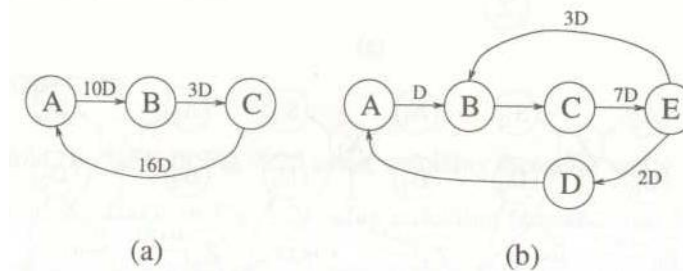
Due date: 2026/6/13

Q1. The DFG shown below describes a 4th order IIR digital filter implemented as cascade of two 2nd order sections. Assume each multiply require 2 u.t. and add requires 1 u.t.

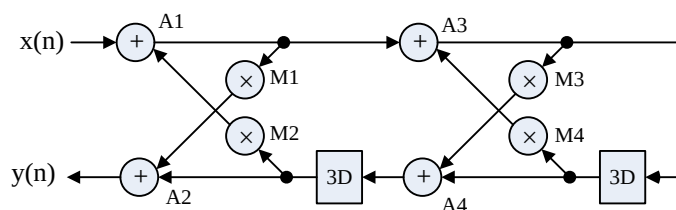
- a) What is the critical path of the DFG? What is the iteration bound of the DFG?
- b) Retime and pipeline the DFG to make the critical path delay equal to iteration bound while keeping the pipeline latency equal to 1, i.e., output $y(n)$ is obtained at clock cycle $n+1$. Note that $y(n)$ of the original design is obtained at clock cycle n .



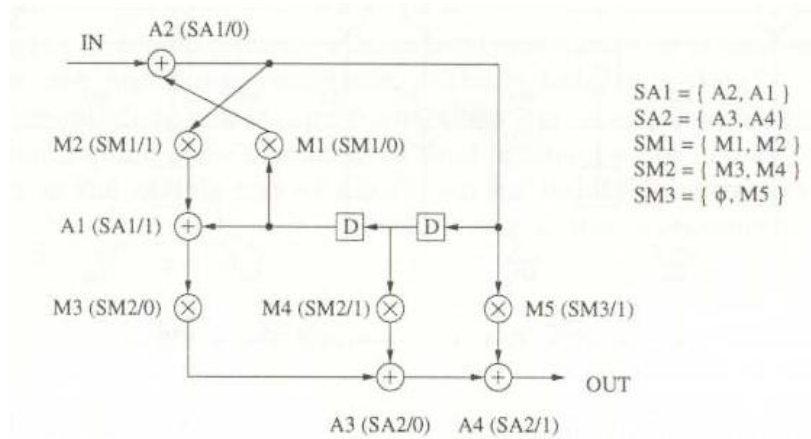
Q2. Unfold the DFG shown below using unfolding factors 2 and 5



Q3. For the DFG shown below, assume the computing times of multiplier and adder are 3 u.t. and 1 u.t., respectively. Unfold the design by a factor of 2 and apply retiming to achieve a critical path delay of 6 u.t..



Q4. For the design shown below.



- For a folded design with 5 folding sets (SA1, SA2, SM1, SM2, SM3) and a folding factor equal to 2, verify if this is a valid folding
- If it is not, find a new valid folding design by modifying the folding orders in each set. **Note:** Each folding set has only two possible folding orders. In total, there are 32 possible combinations of folding orders. You may write a program to enumerate all 32 possible combinations and check if any of them are valid.
- In case, you cannot identify any valid combinations of folding orders in (b), perform retiming first followed by folding
- Derive your folded design without register minimization based on the folding obtained in (b) or (c). You need to draw the folded circuit design.
- Repeat (d) using the forward backward register allocation scheme to minimize the register usage. **Note:** you need to show the derivation details, e.g. life time chart, register allocation table, and so on.